

Simulating and Optimizing Patient Flow using Discrete Event Simulation

Numerical Analysis, Simulation and Modeling Sessional

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Base Paper Overview

Paper Title: *Discrete Event Simulation-Based Analysis and Optimization of Emergency Patient Scheduling Strategies* (Lv et al., 2025)

Domain: Healthcare Management & Operations Research

Brief Summary:

- ▶ Emergency Departments (EDs) face increasing crowding and complexity.
- ▶ Traditional mathematical queuing models fail to capture this complexity due to "state space explosion."
- ▶ The paper constructs a decision support framework using Discrete Event Simulation (DES) to evaluate patient scheduling strategies.
- ▶ **Key Finding:** Dynamic scheduling (Slack-Based Policy) significantly reduces patient wait times compared to static rules (First-Come, First-Served).

Gap in the Base Paper

Base paper does	Base paper does not do
Grid search over the two urgency thresholds (k_1, k_2) only	Optimize physician staffing levels jointly with the thresholds
Compares three heuristic policies (IFP, ALT, SBP)	Compare against a theoretically optimal rule from queueing theory
Brute-force search (hundreds of full simulation runs)	Use a derivative-free numerical optimization method
100 independent replications, no variance reduction	Apply Common Random Numbers to sharpen the comparisons

This is where our project adds genuine novelty rather than re-running the paper's experiments.

Problem Statement & Scope

The Problem: Emergency departments must balance patient satisfaction (low wait times) against financial constraints (idle physician capacity) under unpredictable arrival patterns.

Our Scope & Motive (Methodological Extension):

- ▶ **Rebuild and Validate:** Reconstruct the paper's DES engine in Python and reproduce its headline wait times (46.26 / 37.88 / 35.25 min) within tolerance.
- ▶ **Add a Theoretically Grounded Policy:** Introduce a fourth scheduling rule derived from queueing theory rather than heuristics.
- ▶ **Replace Brute Force with Numerical Optimization:** Tune the policy parameters with a hand-implemented derivative-free optimizer and report its convergence cost.
- ▶ **Optimize Staffing, Not Just Thresholds:** Answer the "right staffing level" question with an explicit economic cost objective.
- ▶ **Strengthen Statistical Evidence:** Apply variance reduction so policy differences are detected with fewer replications.

Our Contribution — Five Work Packages

WP1 — Baseline DES Engine & Replication

Rebuild the five-module simulator (entity definition, event scheduling, resource scheduling, strategy execution, statistical output) and reproduce the paper's results.

WP2 — Theoretically-Grounded Scheduling Rule

Add a fourth policy: the $c\mu$ -rule / Weighted Shortest Expected Processing Time (WSEPT).

WP3 — Numerical Optimizer vs. Grid Search

Replace brute-force search with a hand-coded Nelder–Mead simplex plus penalty functions.

WP4 — Cost-Based Staffing Optimization

Minimize an economic objective over physician counts per shift.

WP5 — Statistical Rigor & Final Comparison

Common Random Numbers, paired t -tests and Cohen's d across all four policies.

Expected Methodology — I. Simulation Engine & Policies

1. Stochastic Modeling (Random Number Generation)

Non-homogeneous Poisson arrivals generated by the *thinning algorithm* from the paper's hourly arrival-rate table, and truncated-exponential consultation times generated by *inverse-CDF sampling*.

2. Discrete Event Simulation (DES) Architecture

An event-driven algorithmic loop that chronologically processes core events (*Patient Arrival*, *Service Start*, *Discharge*) to advance state, rather than ticking second-by-second. The three published policies (IFP, ALT, SBP) are implemented exactly as specified.

3. A Fourth, Provably Motivated Policy

At each decision epoch serve the waiting class maximizing $c_\ell \mu_\ell$, where $\mu_\ell = 1/\mathbb{E}[\text{service time}]$ and $c_\ell = 1/T_\ell$ is a delay-cost weight from the target wait time T_ℓ — optimal for linear holding costs in multi-class queues.

Expected Methodology — II. Optimization & Statistics

4. Derivative-Free Numerical Optimization

A hand-implemented *Nelder–Mead simplex* jointly tunes (k_1, k_2) , with the service-level constraint $SL_\ell \geq SL_{\min}$ handled by a *penalty function*. We benchmark the number of simulation evaluations needed to converge against the paper's two-stage grid search.

5. Cost-Based Staffing Optimization

Minimize

$C(R, k_1, k_2) = \alpha \cdot (\text{physician-hours}) + \beta \cdot (\text{total wait-minutes}) + \gamma \cdot (\text{SLA violation penalty})$
over staffing vectors $R = (R_{\text{morning}}, R_{\text{afternoon}}, R_{\text{night}})$ in a small integer search space.

6. Variance Reduction & Final Comparison

Common Random Numbers reuse identical arrival and service streams across all four policies within each replication, giving narrower confidence intervals and fewer replications for the same statistical power.

Expected Deliverables

- ▶ A from-scratch Python DES simulator reproducing the base paper's reported wait times within statistical tolerance.
- ▶ One clean comparison table over **four** policies (IFP, ALT, SBP, $c\mu$) using the paper's own metrics: mean wait time, service level, utilization, Cohen's d .
- ▶ A measured efficiency figure: simulation evaluations required by Nelder–Mead versus brute-force grid search.
- ▶ A cost-minimizing staffing recommendation per shift, compared against the paper's fixed 5/5/3 physician allocation.
- ▶ Evidence of variance reduction from Common Random Numbers (narrower confidence intervals at equal replication count).